

Instructions: Use proper notation, and show all of your work to receive credit.

1. (2 pts) True or False (circle your answer): The Law of Iterated Expectations says that, for any two random variables X and Y , $E[E(Y|X)] = E(Y)$.

True

False

2. (6 pts) Let Y be a random variable with probability density function given by

$$f(y) = \begin{cases} 2(1-y), & 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}.$$

Find the density function of $U_3 = Y^2$.

3. (4 pts) Suppose that Y has a gamma distribution with $\alpha = n/2$ for some positive integer n and β equal to some specified value. Use the method of moment-generating functions to show that $W = 2Y/\beta$ has a χ^2 distribution with n degrees of freedom.

4. (3 pts) Let $Y_1, Y_2, \dots, Y_n$ be independent and identically distributed random variables such that for $0 < p < 1$, $P(Y_i = 1) = p$ and $P(Y_i = 0) = q = 1 - p$. Find the moment-generating function for Y_1 .

5. (4 pts) If $X \sim N(-2, 3)$ and $Y \sim N(3, 2)$ are independent, find the distribution of $W_1 = X - 2Y$. Be sure to specify the value(s) of the appropriate parameter(s)!

6. (6 pts) Let $Y_1, Y_2, \dots, Y_n$ be independent continuous random variables with cdf $F(y)$ and pdf $f(y)$. Prove that

$$f_{Y_{(n)}}(y) = n[F(y)]^{n-1} \cdot f(y).$$